⚙️ RATCHET IMPROVEMENT REPORT · March 15, 2026

Before & After

Autonomous code improvement run on @ratchet-run/cli

BEFORE		AFTER
83	→ 0 points	83

Issues: **663** → **654** (9 fixed)

4	3	1	20m 34s
Clicks run	Landed clean	Rolled back	Total time

📋 Ratchet ran 4 clicks in 20m 34s, fixing 9 issues. 1 click was rolled back due to test failures.

Issues Fixed

All issue types detected — counts before and after the run

🏆 **Top 3 Wins**

- ↓ **4** lines >120 chars
- ↓ **4** console.log calls in src
- ↓ **1** repeated code lines

ISSUE	SEVERITY	BEFORE	AFTER	CHANGE
lines >120 chars	low	222	**218**	**–4**
console.log calls in src	low	93	**89**	**–4**
repeated code lines	medium	307	**306**	**–1**
functions >50 lines	medium	26	**26**	—
any types	medium	4	**4**	—
await-in-loop patterns	medium	3	**3**	—
missing auth/security controls	medium	1	**1**	—
dead code indicators (TODO, commented code)	low	7	**7**	—

Score Breakdown

Category-by-category improvement

CATEGORY	BEFORE	AFTER	MAX	CHANGE
🧪 Testing	23	23	25	—
🔒 Security	13	13	15	—
📝 Type Safety	15	15	15	—
⚠️ Error Handling	16	16	20	—
⚡ Performance	7	7	10	—
📖 Code Quality	9	9	15	—

before / after

What Ratchet Did

Every change made — 4 clicks, each tested before committing

1 2 3 4

✅ Click 1 Landed

Fixed issues across 8 files

core/utils.ts`, lib/cli.ts`, commands/simulate.ts`, commands/debate.ts` +4 more

↩️ Click 2 Rolled back

Fixed issues across 0 files

✅ Click 3 Landed

Fixed issues across 4 files

lib/cli.ts`, commands/debate.ts`, commands/report.ts`, commands/improve.ts`

✅ Click 4 Landed

Fixed issues across 3 files

commands/scan.ts`, core/scan-cache.ts`, core/scan-constants.ts` (created)

User Impact Assessment

Persona simulation — how real users would experience these changes

OVERALL SENTIMENT

Users could complete the core flow but mid-level technical users hit friction around raw prompt text leaking into commit messages and unreadable swarm output, causing at least one drop-off.

Top Pain Points

- Commit messages contain raw LLM prompt text instead of a human-readable summary of the actual change (Marcus)
- Swarm mode debate output dumps raw JSON to terminal — unreadable mid-run (Marcus)
- Internal error messages like 'simulate confidence below threshold' are opaque to non-expert users (Marcus)
- No clear visual distinction between which swarm agent won and what it changed (Marcus)
- After a rollback, no plain-English explanation of why it failed — just a status code (Marcus)

Top Suggestions

- Fix commit messages immediately — never let prompt text leak into git history; generate a summary of the actual change instead
- Add a clean swarm summary view: 'Agent 2 won this click — changed X, improved score by +3' instead of raw debate JSON
- Translate internal error codes to plain English: 'confidence below threshold' → 'Agents disagreed too much — skipping to stay safe'
- Show a one-line 'what changed' after each click even in non-verbose mode (files touched, score delta)
- Add a --quiet flag that suppresses all agent chatter and shows only the click result line

Critical Drop-offs

- ratchet tighten --pr — Marcus stopped before opening the PR after seeing raw LLM prompt text in git log commit messages

Persona Results

Priya Kapoor (power-user) — neutral

Marcus Chen (casual) — neutral

Dropped at: ratchet tighten --pr — stopped before opening the PR after seeing raw prompt text in git log commit messages

Sofia Rodriguez (new-user) — neutral